

Solving Trigonometric Equations

Student: Anne Takemoto

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Standard: CCSS.MATH.CONTENT.HSF-TF.B.7 — Use inverse functions to solve trigonometric equations that arise in modeling contexts; evaluate the solutions using technology, and interpret them in terms of the context.



Mr. Kenny's Key Point

Two big traps in trig equations:

1. **Always check the range** ($|\sin| \leq 1$, $|\cos| \leq 1$) — extraneous roots like $\cos x = 3$ must be rejected.
2. For **multiple-angle** equations ($2x$, $x/2$, etc.), find ALL solutions in the wider domain first, then translate back to x .

Part A — Direct Equations & Factoring

Directions: Solve each equation on the interval $[0, 2\pi)$ unless stated otherwise. Show all work in the workspace provided.

1. Solve $2 \sin x - 1 = 0$ on $[0, 2\pi)$.

2. Solve $\tan x + 1 = 0$ on $[0, 2\pi)$.

3. Solve $2 \sin \theta \cos \theta + \cos \theta = 0$ on $[0, 2\pi)$.

Hint: Factor out a common trig factor first.

4. Solve $\sin^2 x - \cos^2 x = 0$ on $[0, 2\pi)$.

Hint: Divide both sides by $\cos^2 x$ or use the Pythagorean identity.

Part B — Quadratic in a Trig Function

Directions: Let u = the trig function, factor, then solve. Watch for extraneous roots!

- 5.** Solve $2\cos^2 x - 7\cos x + 3 = 0$ on $[0, 2\pi)$.

Let $u = \cos x$. Factor and reject any root outside $[-1, 1]$.

- 6.** Solve $2\cos^2 \theta + \sin \theta = 1$ on $(0, 2\pi]$.

Use the identity $\cos^2 \theta = 1 - \sin^2 \theta$ to convert to a quadratic in $\sin \theta$.

7. Solve $\tan^2 x = 3$ on $[0, 2\pi)$.

Part C — Multiple-Angle Equations

Directions: For multiple-angle equations, first solve for the compound angle over an extended domain, then convert back to x .

8. Solve $\cos 2x = -1$ on $[0, 2\pi)$.

Let $u = 2x$. Solve for u in $[0, 4\pi)$, then halve.

9. Solve $\sqrt{3} \tan\left(\frac{x}{2}\right) - 1 = 0$ on $[0, 2\pi)$.

Isolate $\tan(x/2)$, solve for $x/2$ in $[0, \pi)$, then double.

10. Solve $\sin 2x = \frac{1}{2}$ on $[0, 2\pi)$.

Let $u = 2x$. Solve for u in $[0, 4\pi)$, then halve. Expect four solutions.
